Pearson Edexcel International A Level · Pure Mathematics 3

Chapter 6 — Differentiation
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P3 第6章 · 微分
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P3-C06-R01 · P3 · Chapter 6

The curve is $y = (x^2 + 1)e^{2x}$.

(a) Find dy/dx in factorised form. (3)

(b) Explain why the curve is strictly increasing for all real x. (3)

(Total for Question 1 is 6 marks)

$f(x) = \dfrac{\ln x}{x}$ for $x > 0$.

(a) Find $f'(x)$. (3)

(b) Find and classify its stationary point. (4)

(Total for Question 2 is 7 marks)

3.

P3-C06-R03 · P3 · Chapter 6

$y = \sin(3x^2)$, where x is in radians.

(a) Find dy/dx and d^2y/dx^2. (4)

(b) Find the tangent at the origin. (3)

(Total for Question 3 is 7 marks)

$C : y = \ln(2x + 1)$, where $x > -1/2$.

(a) Find the equation of the normal at $x = 0$. (3)

(b) Find the point on C where the tangent is parallel to $y = x + 3$. (3)

(Total for Question 4 is 6 marks)

$f(x) = x^2 e^{-x}$ for $x \geq 0$.

(a) Find $f'(x)$ in factorised form. (3)

(b) Find the greatest value on $[0, 4]$ and all points where it is attained. (3)

(Total for Question 5 is 6 marks)

$y = \dfrac{x+1}{x^2+1}$.

(a) Find and simplify dy/dx. (3)

(b) Find the exact stationary x-coordinates and classify them. (4)

(Total for Question 6 is 7 marks)

For $0 < x < \pi/2$, $y = \ln(\sin x)$.

(a) Find dy/dx and d^2y/dx^2. (4)

(b) Find the tangent at $x = \pi/6$. (3)

(Total for Question 7 is 7 marks)

8.

A curve has $y = (ax + b)e^{x}$. At $(0, 3)$ its tangent has gradient 5.

(a) Find a, b. (3)

(b) Find and classify its stationary point. (4)

(Total for Question 8 is 7 marks)

9. P3-C06-R09 · P3 · Chapter 6

$f(x) = \tan(2x)$ for $-\pi/4 < x < \pi/4$.

(a) Find $f'(x)$ and $f''(x)$. (4)

(b) Find the exact point where the tangent has gradient 8 and $x > 0$. (4)

(Total for Question 9 is 8 marks)

$y = \sqrt{4x^2 + 1}$.

(a) Find dy/dx. (3)

(b) Find the exact gradient of the tangent and the normal at $x = 1$. (3)

(Total for Question 10 is 6 marks)

$f(x) = \dfrac{e^x}{x+1}$, with $x \neq -1$.

(a) Show that $f'(x) = xe^x/(x+1)^2$. (3)

(b) Find and classify every stationary point, taking account of the excluded input. (4)

(Total for Question 11 is 7 marks)

$y = x \ln x - x$ for $x > 0$.

(a) Find dy/dx and d^2y/dx^2. (4)

(b) Find the stationary point and show it is a minimum. (4)

(Total for Question 12 is 8 marks)

A model is $P(t) = 30te^{-0.5t}$ for $t \geq 0$.

(a) Find the time and value of its maximum. (4)

(b) Find the rate of change at $t = 4$ and state whether P is increasing or decreasing then. (3)

(Total for Question 13 is 7 marks)

$f(x) = \sin x \cos x$, with x in radians.

(a) Find $f'(x)$ using the product rule, and simplify using a double-angle formula. (3)

(b) Find and classify the stationary points on $[0, \pi]$. (4)

(Total for Question 14 is 7 marks)

$C : y = \ln(x^2 - 4)$ with $x > 2$.

(a) Find dy/dx and d^2y/dx^2. (4)

(b) Show that the gradient is positive and decreasing throughout this domain. (3)

(Total for Question 15 is 7 marks)

$y = (3x - 1)^5$.

(a) Find the first two derivatives. (4)

(b) Find the normal at $x = 0$. (3)

(Total for Question 16 is 7 marks)

$f(x) = \dfrac{x^2}{2x+1}$, with $x \neq -1/2$.

(a) Find $f'(x)$ in factorised form. (3)

(b) Find all stationary points and classify them. (4)

(Total for Question 17 is 7 marks)

$C : y = e^{3x} - 6e^{x}$.

(a) Find its stationary point exactly. (4)

(b) Use the second derivative to classify it. (3)

(Total for Question 18 is 7 marks)

$y = \ln\left(\dfrac{x+1}{x-1}\right)$ for $x > 1$.

(a) Find dy/dx in simplest form. (3)

(b) Find the exact point where the tangent has gradient $-1/4$. (3)

(Total for Question 19 is 6 marks)

$f(x) = \cos^2(2x)$ for $0 \le x \le \pi/2$.

(a) Find $f'(x)$ and express it using a single sine function. (3)

(b) Find all points on the interval with horizontal tangents, and state their function values. (3)

(Total for Question 20 is 6 marks)

21.

A curve is $y = (x - a)e^{2x}$ and has a stationary point at $x = 1$.

(a) Find a. (3)

(b) Find the coordinates of the stationary point and prove it is a minimum. (4)

(Total for Question 21 is 7 marks)

$f(x) = \dfrac{\sin x}{1 + \cos x}$ wherever defined.

(a) Show that $f'(x) = 1/(1 + \cos x)$. (3)

(b) Find the tangent at $x = \pi/2$. (3)

(Total for Question 22 is 6 marks)

$C : y = \ln(1 + e^x)$.

(a) Find the first two derivatives. (4)

(b) Find the tangent at $x = 0$ and explain why the gradient is increasing for every real x. (4)

(Total for Question 23 is 8 marks)

$y = xe^{x^2}$.

(a) Find dy/dx in factorised form. (3)

(b) Find d^2y/dx^2 and state all real inputs where this derivative is zero. (4)

(Total for Question 24 is 7 marks)

For $x > 0$, $f(x) = \ln x + 2/x$.

(a) Find and classify the stationary point. (4)

(b) Find the greatest and least values of f on $[1, 4]$. (3)

(Total for Question 25 is 7 marks)

$C : y = \sin x + \cos x$ on $0 \leq x < 2\pi$.

(a) Find its stationary points exactly. (4)

(b) Use the second derivative to classify them and compare with the auxiliary-angle form. (3)

(Total for Question 26 is 7 marks)

$y = (\ln x)^2$ for $x > 0$.

(a) Find the first and second derivatives. (4)

(b) Find the stationary point and all inputs where $d^2y/dx^2 = 0$. (3)

(Total for Question 27 is 7 marks)

$C : y = \dfrac{x}{\sqrt{x^2 + 1}}$.

(a) Show that $dy/dx = (x^2 + 1)^{-3/2}$. (4)

(b) Find the exact gradient at $x = \sqrt{3}$ and the normal there. (3)

(Total for Question 28 is 7 marks)

29. P3-C06-R29 · P3 · Chapter 6

The curve C is $y = \ln(5 - 2x)$ with $x < 5/2$.

(a) Find the tangent at the point where the curve meets the x-axis. (3)

(b) Find where this tangent meets the vertical asymptote of C. (3)

(Total for Question 29 is 6 marks)

$y = e^{\sin x}$ for $0 \leq x \leq 2\pi$.

(a) Find dy/dx and d^2y/dx^2. (4)

(b) Find and classify the stationary points on the interior of the interval. (4)

(Total for Question 30 is 8 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P3-C06-R01 · 6 marks

(a) · 3 marks

$2e^{2x}(x^2+x+1)$.

乘积法则给2xe^{2x}+2(x²+1)e^{2x}，提出2e^{2x}。

M1	乘积法则
M1	指数链式因子2
A1	因式形式

(b) · 3 marks

$x^2+x+1=(x+1/2)^2+3/4>0$ and $e^{2x}>0$, so $y'>0$.

配方证明二次因子恒正，结合指数恒正，导数严格正。

M1	二次配方
M1	两因子正性
A1	单调结论

Question 2

P3-C06-R02 · 7 marks

(a) · 3 marks

$(1-\ln x)/x^2$.

商法则[(1/x)x−lnx]/x²；也可乘x^{-1}。

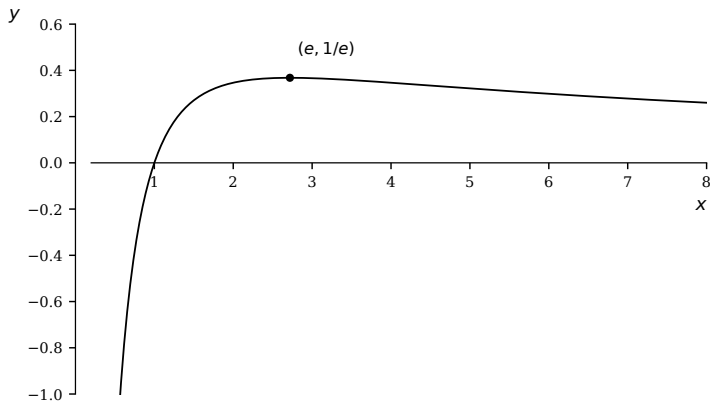
M1	商法则
M1	正确ln导数
A1	简式

(b) · 4 marks

$(e, 1/e)$ is a maximum.

导数分母正，分子1−lnx在e前正、后负，因此由增转减。

M1	驻点方程
M1	正确纵坐标
M1	导数符号改变
A1	分类



Suggested sketch

Question 3

P3-C06-R03 · 7 marks

(a) · 4 marks

$6x\cos(3x^2);\ 6\cos(3x^2) - 36x^2\sin(3x^2).$

一阶用链式，二阶对6x和余弦乘积求导，内层导数6x再次出现。

M1	一阶链式
A1	一阶导
M1	二阶乘积链式
A1	二阶导

(b) · 3 marks

$y = 0.$

原函数在0为0，导式在0也为0，所以切线为x轴。

B1	点
M1	梯度代值
A1	切线

Question 4

P3-C06-R04 · 6 marks

(a) · 3 marks

$y = -x/2$.

原点在曲线上，切线斜率2/(2x+1)在0为2，法线−1/2。

M1	链式对数导数
M1	法线负倒数
A1	方程

(b) · 3 marks

$(1/2, \ln 2)$.

令2/(2x+1)=1得x1/2，回代。

M1	梯度条件
M1	域及回代
A1	点

Question 5

P3-C06-R05 · 6 marks

(a) · 3 marks

$e^{-x}x(2-x)$.

乘积法则，e^{-x}导数带负号，提出e^{-x}x。

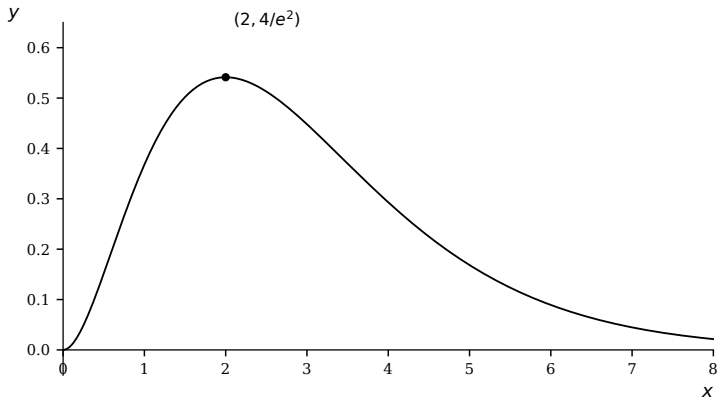
M1	乘积法则
M1	负链式因子
A1	因式式

(b) · 3 marks

$4e^{-2}$, only at $x=2$.

导式在0至2正、2至4负；比较端点0与16e^{-4}，驻点最大。

M1	候选点和端点
M1	符号或比较
A1	最大值和唯一位置



Suggested sketch

Question 6

P3-C06-R06 · 7 marks

(a) · 3 marks

$(-x^2-2x+1)/(x^2+1)^2$.

商法则分子$(x^2+1)-2x(x+1)$，分母平方始终正。

M1	正确商法则
M1	分子化简
A1	导式

(b) · 4 marks

$x=-1-\sqrt{2}$ minimum; $x=-1+\sqrt{2}$ maximum.

分子$2-(x+1)^2$，在两根间正，外侧负，所以左根由负转正，右根由正转负。

M1	两根
M1	导数符号
M1	类型变化
A1	分类

Question 7

P3-C06-R07 · 7 marks

(a) · 4 marks

$\cot x;\ -\operatorname{cosec}^2 x.$

ln链式导数cos/sin=cot，再用余切标准导数或商法则。

M1	对数链式
A1	一阶
M1	二阶求导
A1	二阶

(b) · 3 marks

$y=\sqrt{3}(x-\pi/6)-\ln 2.$

y=ln(1/2)=−ln2，梯度cotπ/6=√3。

B1	纵坐标
M1	梯度
A1	切线

Question 8

P3-C06-R08 · 7 marks

(a) · 3 marks

$b = 3, a = 2.$

点给b3；y'=(ax+a+b)e^x，在0给a+b5。

B1	b
M1	乘积求导与梯度条件
A1	a

(b) · 4 marks

$(-5/2, -2e^{-5/2})$ is a minimum.

导式(2x+5)e^x，零点−5/2，分子线性由负转正；原线性因子2x+3在此为−2。

M1	导数零点
M1	正确纵坐标
M1	符号变化
A1	分类

Question 9

P3-C06-R09 · 8 marks

(a) · 4 marks

$2\sec^2(2x)$; $8\sec^2(2x)\tan(2x)$.

一阶有输入导数2；二阶对$\sec^2$复合求导再乘2。

M1	一阶链式
A1	一阶
M1	二阶链式
A1	二阶

(b) · 4 marks

$(\pi/6, \sqrt{3})$.

$2\sec^2 2x=8$给$\cos^2 2x\,1/4$；输入$2x\in(0,\pi/2)$，取$\cos 1/2$即$2x\pi/3$。

M1	梯度方程
M1	输入范围选支
M1	原函数回代
A1	点

Question 10

P3-C06-R10 · 6 marks

(a) · 3 marks

$4x/\sqrt{4x^2+1}$.

写成(4x^2+1)^{1/2}，外导1/2乘内导8x。

M1	外幂导数
M1	内导链式
A1	简式

(b) · 3 marks

Tangent $4/\sqrt{5}$; normal $-\sqrt{5}/4$.

代导式x1，再取负倒数。

M1	导数代值
M1	法线负倒数
A1	两比

Question 11

P3-C06-R11 · 7 marks

(a) · 3 marks

Quotient numerator $e^x(x+1) - e^x = xe^x$.

商法则后公因子提取，分母必须保留平方。

M1	正确商法则
M1	分子化简
A1	目标式

(b) · 4 marks

Only $(0, 1)$, a minimum on the branch $x > -1$.

指数不为零，分母正，导数符号由x；在0负转正。x=−1未定义，不能算驻点或连接两支比较全局。

M1	找唯一导数零点
M1	域分支说明
M1	符号改变
A1	分类

Question 12

P3-C06-R12 · 8 marks

(a) · 4 marks

$\ln x$; $1/x$.

xlnx乘积导数lnx+1，减去x导数1后抵消。

M1	乘积导数
A1	一阶
M1	再求导
A1	二阶

(b) · 4 marks

$(1,-1)$; $y''(1)=1>0$.

lnx0给x1，回代−1；二阶正。

M1	解驻点
M1	纵坐标
M1	二阶判别
A1	点与类型

Question 13

P3-C06-R13 · 7 marks

(a) · 4 marks

$t = 2$; maximum $60/e$.

P'=30e^{−.5t}(1−.5t)，在2由正转负，代P2。

M1	乘积链式
M1	驻点
M1	最大性
A1	时间和值

(b) · 3 marks

$P'(4) = -30e^{-2}$; decreasing.

代导式得负值，负号说明此时下降。

M1	正确导式代值
A1	精确变化率
B1	方向解释

Question 14

P3-C06-R14 · 7 marks

(a) · 3 marks

$\cos^2 x - \sin^2 x = \cos 2x$.

分别求导sin与cos，第二项带负；再用双角恒等式。

M1	乘积法则
M1	正确负号
A1	双角式

(b) · 4 marks

$(\pi/4, 1/2)$ maximum; $(3\pi/4, -1/2)$ minimum.

cos2x0得两根，f″=−2sin2x在两处负、正。

M1	区间内导数零点
M1	纵坐标
M1	二阶符号
A1	分类

Question 15

P3-C06-R15 · 7 marks

(a) · 4 marks

$2x/(x^2-4)$; $-2(x^2+4)/(x^2-4)^2$.

一阶ln链式，二阶用商法则，分子$2(x^2-4)-4x^2$。

M1	一阶链式
A1	一阶
M1	二阶商法则
A1	简式

(b) · 3 marks

$y'>0$ and $y''<0$ for $x>2$.

一阶分子分母都正；二阶分子严格负、分母正，分别说明曲线上升且其斜率下降。

M1	一阶符号
M1	二阶符号
A1	区分两结论

Question 16

P3-C06-R16 · 7 marks

(a) · 4 marks

$15(3x-1)^4$; $180(3x-1)^3$.

两次外幂求导都乘输入导数3，第二次15×4×3=180。

M1	一阶链式
A1	一阶
M1	二阶链式
A1	二阶

(b) · 3 marks

$y=-x/15-1$.

点(0,−1)，切线梯度15，法线−1/15。

B1	点
M1	法线斜率
A1	方程

Question 17

P3-C06-R17 · 7 marks

(a) · 3 marks

$2x(x+1)/(2x+1)^2$.

商法则分子2x(2x+1)−2x²=2x(x+1)。

M1	商法则和内导2
M1	因式分解
A1	导式

(b) · 4 marks

$(-1,-1)$ maximum; $(0,0)$ minimum.

导数分母正，x(x+1)在−1两侧正转负，在0负转正；注意−1/2将域分开。

M1	两导数零点
M1	纵坐标
M1	各分支符号
A1	分类

Question 18

P3-C06-R18 · 7 marks

(a) · 4 marks

$\left(\frac{1}{2}\ln 2, -4\sqrt{2}\right)$.

y'=3e^x(e^{2x}−2)，零点e^{2x}=2，原式e^x(2−6)。

M1	因式化导数
M1	对数求x
M1	精确y
A1	点

(b) · 3 marks

Minimum, since $y'' = 9e^{3x} - 6e^{x}$ has value $12\sqrt{2} > 0$ there.

e^{3x}=2√2，代9×2√2−6√2=12√2。

M1	正确二阶导
M1	精确代值
A1	最小性

Question 19

P3-C06-R19 · 6 marks

(a) · 3 marks

$-2/(x^2-1)$.

先写ln(x+1)−ln(x−1)，导数1/(x+1)−1/(x−1)再通分。

M1	合法拆对数
M1	两项求导
A1	简式

(b) · 3 marks

$(3, \ln 2)$.

−2/(x²−1)=−1/4给x²9，域x>1只取3；回代ln(4/2)。

M1	梯度方程
M1	域筛选
A1	点

Question 20

P3-C06-R20 · 6 marks

(a) · 3 marks

$-2\sin 4x$.

链式给−4cos2xsin2x，用sin4x=2sin2xcos2x。

M1	外平方与内角链式
M1	双角化简
A1	导式

(b) · 3 marks

$x = 0, \pi/4, \pi/2$; values $1, 0, 1$.

4x在0到2π，sin4x0的输入0、π、2π；端点的导函数也为0。

M1	输入范围零点
M1	包括端点
A1	三x及值

Question 21

P3-C06-R21 · 7 marks

(a) · 3 marks

$a = 3/2$.

y'=e^{2x}[1+2(x−a)]，在1零给3−2a=0。

M1	乘积链式
M1	驻点条件
A1	a

(b) · 4 marks

$(1, -e^2/2)$; the derivative factor is $2(x-1)$ and changes from negative to positive.

代a得原函数(1−1.5)e^2；指数正，导数由线性因子定号。

M1	纵坐标
M1	因式符号
M1	负转正
A1	类型

Question 22

P3-C06-R22 · 6 marks

(a) · 3 marks

Quotient numerator $\cos x(1+\cos x)+\sin^2 x = 1+\cos x$.

分母导数为$-\sin$，商法则中的相减再变加；用平方和约分。

M1	正确商法则符号
M1	平方和化简
A1	导式

(b) · 3 marks

$y = x + 1 - \pi/2$.

点$(\pi/2, 1)$，导数1，点斜式。

B1	点
M1	梯度
A1	方程

Question 23

P3-C06-R23 · 8 marks

(a) · 4 marks

$e^x/(1+e^x)$; $e^x/(1+e^x)^2$.

一阶对数链式，二阶商法则两e^{2x}项抵消。

M1	一阶链式
A1	一阶
M1	二阶商法则
A1	二阶

(b) · 4 marks

$y = x/2 + \ln 2$; second derivative is strictly positive.

点(0,ln2)，一阶1/2；二阶指数正、分母正，梯度递增。

M1	点与斜率
A1	方程
M1	二阶符号
A1	梯度说明

Question 24

P3-C06-R24 · 7 marks

(a) · 3 marks

$e^{x^2}(1+2x^2)$.

乘积导数e^{x^2}+x e^{x^2}2x，提出指数。

M1	乘积法则
M1	内层2x
A1	因式形式

(b) · 4 marks

$2xe^{x^2}(3+2x^2)$; only $x=0$.

再乘积求导，合并为e^{x^2}(6x+4x^3)；指数和3+2x^2恒正，所以仅x0。

M1	二阶乘积链式
M1	因式整理
M1	正因子判零
A1	导式及根

Question 25

P3-C06-R25 · 7 marks

(a) · 4 marks

$(2, \ln 2 + 1)$ minimum.

f′=(x−2)/x²，在2负转正，原值ln2+1。

M1	正确导数
M1	零点与原值
M1	符号变化
A1	类型

(b) · 3 marks

Least $1 + \ln 2$ at 2; greatest 2 at 1.

f1=2，f4=ln4+1/2≈1.8863，比较端点和驻点。

M1	两端点数值
M1	与驻点比较
A1	两全局极值

Question 26

P3-C06-R26 · 7 marks

(a) · 4 marks

$(\pi/4, \sqrt{2})$ and $(5\pi/4, -\sqrt{2})$.

y'=cosx−sinx=0，tan1但需cos非零，得到两角并代原式。

M1	正确导数
M1	全部角
M1	精确y
A1	两点

(b) · 3 marks

Maximum then minimum; $y'' = -y$ and $y = \sqrt{2}\sin(x + \pi/4)$.

二阶在两个点分别−√2和√2；辅助角幅值√2，与最高最低一致。

M1	二阶符号分类
M1	辅助角式
A1	一致说明

Question 27

P3-C06-R27 · 7 marks

(a) · 4 marks

$2\ln x/x$; $2(1-\ln x)/x^2$.

一阶链式2lnx乘1/x，二阶用商法则或负幂乘积。

M1	一阶链式
A1	一阶
M1	二阶商法则
A1	二阶

(b) · 3 marks

Stationary $(1,0)$; second derivative zero at $x=e$.

lnx0给1，另一方程1−lnx0给e；这两条件不能混淆。

M1	一阶导零点
M1	二阶导零点
A1	点与输入

Question 28

P3-C06-R28 · 7 marks

(a) · 4 marks

Write $y = x(x^2+1)^{-1/2}$; product derivative has numerator $(x^2+1) - x^2 = 1$.

乘积与链式后以(x²+1)^{3/2}为公分母，分子抵消。

M1	幂式乘积
M1	链式内导
M1	分子化简
A1	目标式

(b) · 3 marks

Gradient $1/8$; normal $y - \sqrt{3}/2 = -8(x - \sqrt{3})$.

分母(3+1)^{3/2}=8，点(√3,√3/2)，法线负倒数−8。

M1	导数精确代值
M1	点与法线斜率
A1	方程

Question 29

P3-C06-R29 · 6 marks

(a) · 3 marks

$y = -2x + 4.$

y0给5−2x1，点(2,0)；导数−2/(5−2x)，在2为−2。

M1	截距点
M1	链式斜率
A1	切线

(b) · 3 marks

$(5/2, -1).$

渐近线x=5/2，与切线联立纵坐标−5+4=−1；该点不在原曲线上。

B1	渐近线
M1	代切线
A1	交点

Question 30

P3-C06-R30 · 8 marks

(a) · 4 marks

$e^{\sin x}\cos x$; $e^{\sin x}(\cos^2 x - \sin x)$.

一阶指数链式，二阶乘积法则，cos导数−sin。

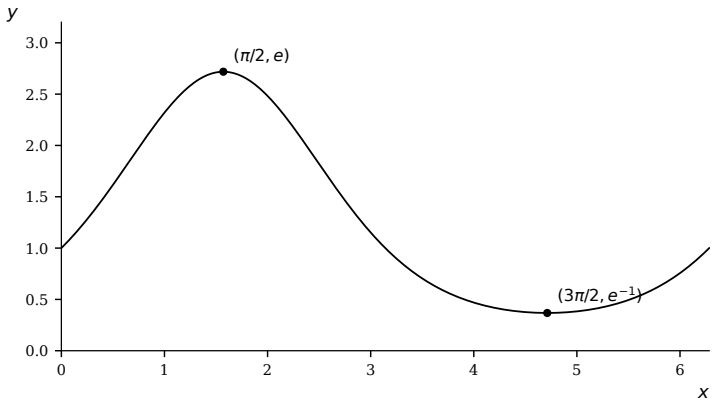
M1	一阶链式
A1	一阶
M1	二阶乘积
A1	二阶

(b) · 4 marks

$(\pi/2, e)$ maximum; $(3\pi/2, e^{-1})$ minimum.

指数不零，cos零给两内部点；二阶在前为−e、后为e^{-1}。

M1	导数零点
M1	精确纵坐标
M1	二阶符号
A1	分类



Suggested sketch